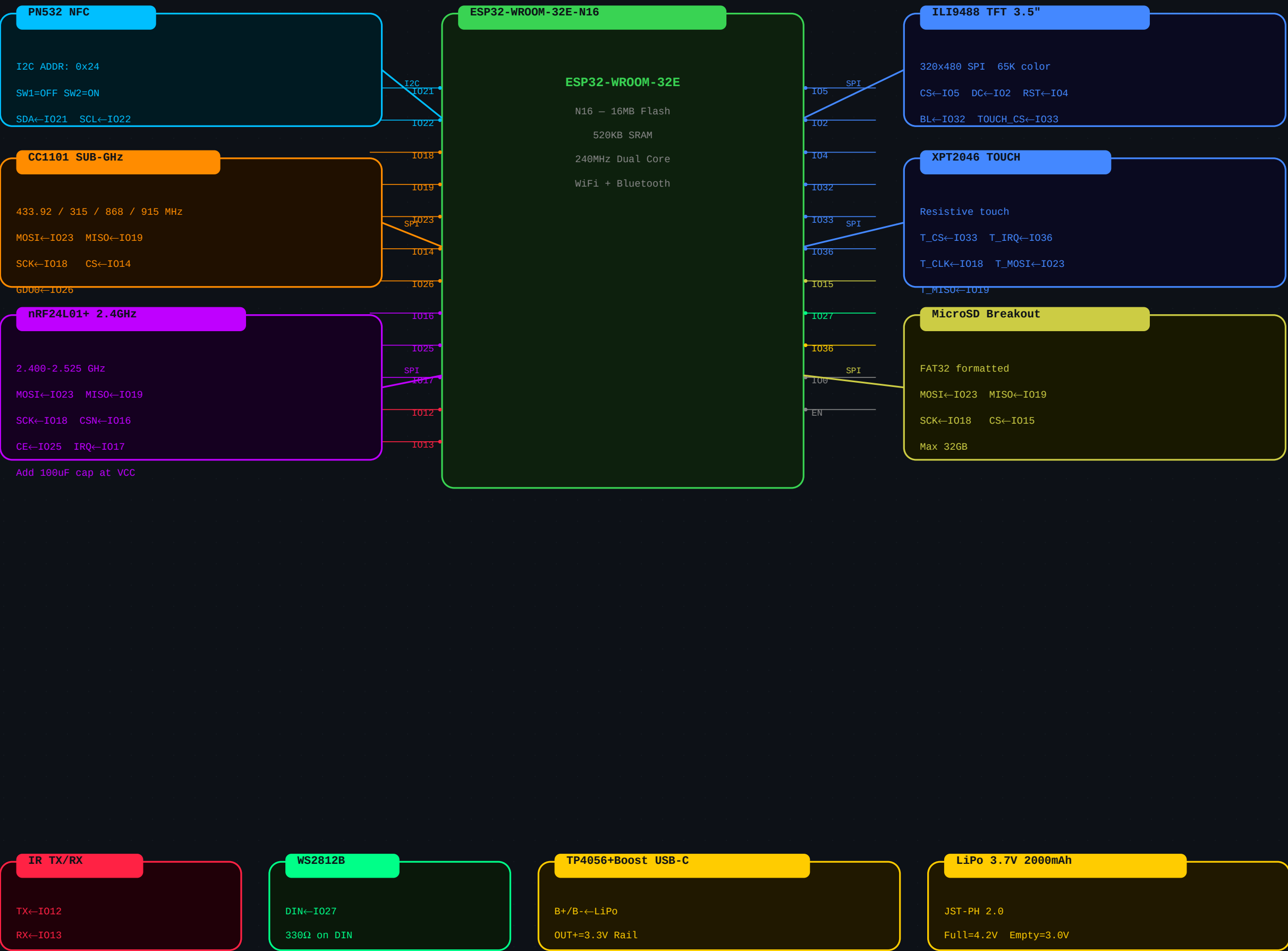


SIGNAL GOBLIN

ESP32-WROOM-32E-N16 | ILI9488 3.5" | PN532 / CC1101 / nRF24 / IR

SYSTEM OVERVIEW

All modules, buses, and power rail at a glance



DETAILED MODULE CONNECTIONS

Every pin, every wire

SPI BUS – SHARED BY 5 DEVICES

|                                               |          |                               |
|-----------------------------------------------|----------|-------------------------------|
| GPI018                                        | SCK →    | CC1101, nRF24, TFT, SD, Touch |
| GPI019                                        | MISO ←   | CC1101, nRF24, TFT, SD, Touch |
| GPI023                                        | MOSI →   | CC1101, nRF24, TFT, SD, Touch |
|                                               |          |                               |
| GPI05                                         | CS → LOW | TFT display ONLY              |
| GPI014                                        | CS → LOW | CC1101 ONLY                   |
| GPI016                                        | CS → LOW | nRF24 ONLY                    |
| GPI015                                        | CS → LOW | SD card ONLY                  |
| GPI033                                        | CS → LOW | Touch XPT2046 ONLY            |
| ! NEVER enable two CS pins LOW simultaneously |          |                               |

I2C BUS

|         |          |                           |
|---------|----------|---------------------------|
| GPI021  | SDA      | PN532 pin 3 (addr 0x24)   |
| GPI022  | SCL      | PN532 pin 4               |
| R1 4.7k | SDA→3.3V | Pull-up required          |
| R2 4.7k | SCL→3.3V | Pull-up required          |
| SW1     | = OFF    | PN532 I2C mode            |
| SW2     | = ON     | PN532 I2C mode ← MUST SET |

nRF24L01+

|      |        |              |
|------|--------|--------------|
| VCC  | 3.3V   | + 100uF cap! |
| GND  | GND    |              |
| CE   | GPI025 | Chip enable  |
| CSN  | GPI016 | Chip select  |
| SCK  | GPI018 | SPI clock    |
| MOSI | GPI023 | SPI data     |
| MISO | GPI019 | SPI data     |
| IRQ  | GPI017 | Interrupt    |

MicroSD

|      |        |  |
|------|--------|--|
| VCC  | 3.3V   |  |
| GND  | GND    |  |
| MOSI | GPI023 |  |
| MISO | GPI019 |  |
| SCK  | GPI018 |  |
| CS   | GPI015 |  |

ILI9488 3.5" DISPLAY

|          |        |                  |
|----------|--------|------------------|
| VCC      | 3.3V   | Power            |
| GND      | GND    | Ground           |
| CS       | GPI05  | Chip select      |
| RESET    | GPI04  | Active LOW reset |
| DC/RS    | GPI02  | Data/Command     |
| SDI/MOSI | GPI023 | SPI data         |
| SCK      | GPI018 | SPI clock        |

XPT2046 TOUCH (on TFT board)

|                                   |        |                   |
|-----------------------------------|--------|-------------------|
| T_CS                              | GPI033 | Touch chip select |
| T_IRQ                             | GPI036 | Touch interrupt   |
| T_CLK                             | GPI018 | SPI clock shared  |
| T_MOSI                            | GPI023 | SPI MOSI shared   |
| T_MISO                            | GPI019 | SPI MISO shared   |
| Note: GPIO36 input-only no pullup |        |                   |

CC1101 SUB-GHz

|      |        |                 |
|------|--------|-----------------|
| VCC  | 3.3V   | Power           |
| GND  | GND    |                 |
| MOSI | GPI023 | SPI data in     |
| MISO | GPI019 | SPI data out    |
| SCK  | GPI018 | SPI clock       |
| CSN  | GPI014 | Chip select     |
| CSRX | GPI026 | Data/IRQ output |

IR TX/RX

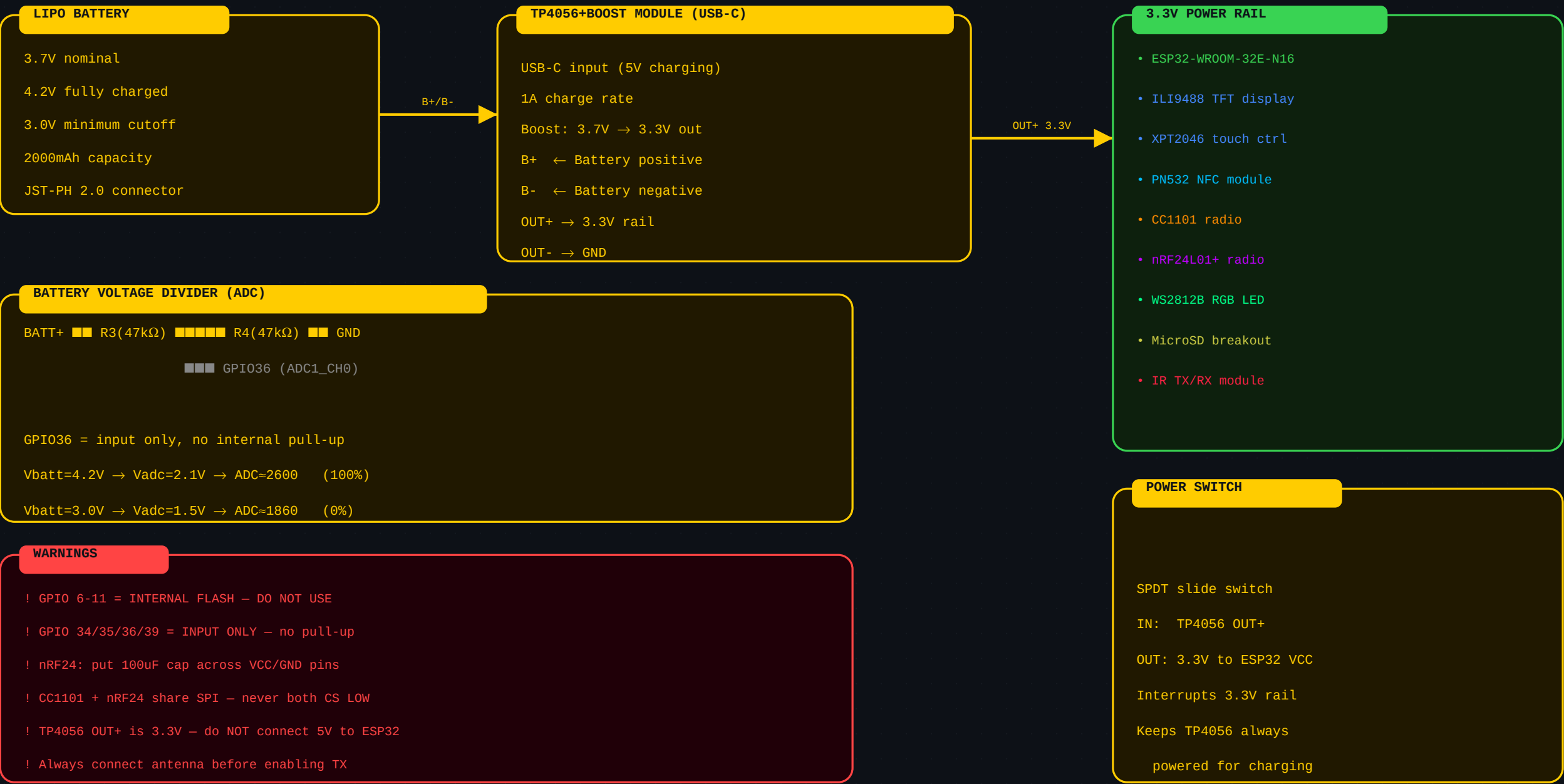
|     |        |                |
|-----|--------|----------------|
| VCC | 3.3V   |                |
| GND | GND    |                |
| TX  | GPI012 | 38kHz PWM out  |
| RX  | GPI013 | 38kHz demod in |

WS2812B

|      |        |               |
|------|--------|---------------|
| VDD  | 3.3V   |               |
| VSS  | GND    |               |
| DIN  | GPI027 | 330Ω series R |
| DOUT | N/C    | chain if >1   |

POWER SYSTEM

TP4056+Boost module, LiPo battery, 3.3V rail



COMPLETE GPIO REFERENCE

Every ESP32-WROOM-32E-N16 pin assionment

| GPIO | FUNCTION | MODULE  | DIR | NOTES                        |
|------|----------|---------|-----|------------------------------|
| I00  | B00T     | System  | IN  | Hold GND to enter flash mode |
| I01  | TX0      | UART0   | OUT | Serial debug / programming   |
| I02  | TFT_DC   | ILI9488 | OUT | Data/Command select          |
| I03  | RX0      | UART0   | IN  | Serial debug / programming   |
| I04  | TFT_RST  | ILI9488 | OUT | Display reset active LOW     |
| I05  | TFT_CS   | ILI9488 | OUT | Display SPI chip select      |
| I06  | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I07  | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I08  | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I09  | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I010 | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I011 | ■ FLASH  | SYSTEM  | --- | INTERNAL FLASH – DO NOT USE  |
| I012 | IR_TX    | IR      | OUT | IR transmit 38kHz PWM        |
| I013 | IR_RX    | IR      | IN  | IR receive demodulated       |
| I014 | CC_CS    | CC1101  | OUT | CC1101 SPI chip select       |
| I015 | SD_CS    | MicroSD | OUT | SD card SPI chip select      |
| I016 | NRF_CSN  | nRF24   | OUT | nRF24 chip select            |
| I017 | NRF_IRQ  | nRF24   | IN  | nRF24 interrupt active LOW   |
| I018 | SPI_SCK  | SPI BUS | OUT | Shared: TFT+CC+nRF+SD+Touch  |
| I019 | SPI_MISO | SPI BUS | IN  | Shared: TFT+CC+nRF+SD+Touch  |
| I021 | I2C_SDA  | I2C BUS | I/O | PN532 NFC data line          |
| I022 | I2C_SCL  | I2C BUS | OUT | PN532 NFC clock line         |
| I023 | SPI_MOSI | SPI BUS | OUT | Shared: TFT+CC+nRF+SD+Touch  |
| I025 | NRF_CE   | nRF24   | OUT | nRF24 chip enable            |
| I026 | CC_GD00  | CC1101  | IN  | CC1101 data output 0         |
| I027 | RGB_DIN  | WS2812B | OUT | RGB LED data (330Ω series)   |
| I032 | TFT_BL   | ILI9488 | OUT | Display backlight PWM        |
| I033 | TOUCH_CS | XPT2046 | OUT | Touch SPI chip select        |
| I034 | N/C      | ---     | IN  | Input only – no pull-up      |
| I035 | N/C      | ---     | IN  | Input only – no pull-up      |
| I036 | BATT_ADC | ADC     | IN  | Battery divider + Touch IRQ  |
| I039 | N/C      | ---     | IN  | Input only – no pull-up      |
| EN   | RESET    | System  | IN  | Active LOW chip reset        |
| 3V3  | POWER IN | Rail    | PWR | 3.3V from TP4056 OUT+        |
| GND  | GROUND   | Rail    | PWR | Common ground                |

BREADBOARD BUILD GUIDE

Step by step – power on first, modules one at a time

STEP 1 – POWER UP

- TP4056 OUT+ → ESP32 3V3 pin
- TP4056 OUT- → ESP32 GND pin
- LiPo (+) → TP4056 B+ pad
- LiPo (-) → TP4056 B- pad
- VERIFY: multimeter reads ~3.3V at ESP32 3V3/GND
- Power switch wired in series on 3.3V line

STEP 2 – SSD1306 OLED (if testing now)

- OLED VCC → 3.3V rail
- OLED GND → GND rail
- OLED SDA → GPIO21
- OLED SCL → GPIO22
- 4.7kΩ resistor: SDA → 3.3V
- 4.7kΩ resistor: SCL → 3.3V
- Run I2C scanner – should see 0x3C

STEP 3 – PN532 NFC

- SET SW1=OFF, SW2=ON before wiring!
- PN532 VCC → 3.3V
- PN532 GND → GND
- PN532 SDA → GPIO21
- PN532 SCL → GPIO22
- I2C scanner should show 0x24

STEP 4 – CC1101 Sub-GHz

- CC1101 VCC → 3.3V
- CC1101 GND → GND
- CC1101 MOSI → GPIO23
- CC1101 MISO → GPIO19
- CC1101 SCK → GPIO18
- CC1101 CSN → GPIO14
- CC1101 GD00 → GPIO26
- Attach SMA antenna before any TX!

STEP 5 – nRF24L01+

- Solder 100uF cap across VCC/GND on module
- nRF24 VCC → 3.3V
- nRF24 GND → GND
- nRF24 CE → GPIO25
- nRF24 CSN → GPIO16
- nRF24 SCK → GPIO18 (shared)
- nRF24 MOSI → GPIO23 (shared)
- nRF24 MISO → GPIO19 (shared)
- nRF24 IRQ → GPIO17

STEP 6 – ILI9488 TFT Display

- TFT VCC → 3.3V
- TFT GND → GND
- TFT CS → GPIO5
- TFT RESET → GPIO4
- TFT DC → GPIO2
- TFT MOSI → GPIO23 (shared)
- TFT SCK → GPIO18 (shared)
- TFT LED → GPIO32
- TFT MISO → GPIO19 (shared)
- T\_CS → GPIO33
- T\_IRQ → GPIO36

STEP 7 – IR + RGB + SD

- IR VCC → 3.3V    IR GND → GND
- IR TX → GPIO18    IR RX → GPIO19